CHAPTER 3: SYSTEM ARCHITECTURE AND BLOCK DIAGRAM

3.1 Introduction

The architecture of an IoT-based Smart Irrigation and Plant Monitoring System defines how different hardware and software components interact to perform monitoring and irrigation tasks. The proposed system integrates environmental sensors, a microcontroller, display devices, and irrigation control mechanisms to provide an automated solution for plant monitoring.

The primary objective of the system architecture is to collect environmental data, process sensor readings, display information, and control irrigation automatically whenever required.

The system follows a layered architecture consisting of:

1. Input Layer
2. Processing Layer
3. Output Layer
4. Irrigation Control Layer

Each layer performs specific functions to ensure reliable and efficient operation.

3.2 Overall System Architecture

The complete system consists of the following major components:

Input Devices

- DHT11 Temperature and Humidity Sensor
- Soil Moisture Sensor

Processing Unit

- ESP32 Development Board

Output Devices

- OLED Display

Control Devices

- Relay Module
- Water Pump

The sensors continuously collect environmental information and send it to the ESP32. The ESP32 processes the received data and displays the results on the OLED screen. Based on predefined moisture thresholds, the controller decides whether irrigation is required and controls the relay accordingly.

3.3 Block Diagram

```
                 +----------------------+
                 |     DHT11 Sensor     |
                 | Temperature/Humidity |
                 +----------+-----------+
                            |
                            |
                            v

                 +----------------------+
                 | Soil Moisture Sensor |
                 +----------+-----------+
                            |
                            |
                            v

          +---------------------------------------+
          |                ESP32                  |
          |                                       |
          | • Reads Sensor Data                   |
          | • Calculates Moisture Percentage      |
          | • Controls Relay Module               |
          | • Updates OLED Display                |
          +---------+-----------------------------+
                    |
        +-----------+------------+
        |                        |
        v                        v

+--------------------+   +--------------------+
|    OLED Display    |   |    Relay Module    |
|                    |   |                    |
| Temperature        |   | Switches Pump      |
| Humidity           |   | ON/OFF             |
| Soil Moisture      |   +---------+----------+
| Pump Status        |             |
+--------------------+             |
                                   v

                         +---------------+
```

```
| Water Pump    |
+-------+-------+
        |
        v

+---------------+
| Irrigation    |
| of Plants     |
+---------------+
```

3.4 Input Layer

The input layer consists of sensors responsible for collecting environmental information.

3.4.1 DHT11 Sensor

The DHT11 sensor measures:

- Temperature
- Relative Humidity

The sensor provides digital output which is directly read by the ESP32 through GPIO4.

Specifications

- Operating Voltage: 3.3V–5V
- Temperature Range: 0°C to 50°C
- Humidity Range: 20% to 90%
- Digital Communication Interface

Purpose

The DHT11 helps monitor environmental conditions affecting plant growth.

3.4.2 Soil Moisture Sensor

The soil moisture sensor measures water content present in soil.

Working Principle

The sensor determines moisture levels by measuring soil conductivity.

- Wet soil → Higher conductivity

- Dry soil → Lower conductivity

The analog output is connected to GPIO34 of ESP32.

Specifications

- Operating Voltage: 3.3V–5V
- Analog Output
- Low Power Consumption

Purpose

The sensor provides the information required to determine whether irrigation is necessary.

3.5 Processing Layer

3.5.1 ESP32 Development Board

The ESP32 acts as the central processing unit of the system.

Features

- Dual-Core Processor
- Built-in Wi-Fi
- Built-in Bluetooth
- Multiple GPIO Pins
- ADC Channels
- Low Power Consumption

Functions Performed

1. Reads sensor data.
2. Processes moisture values.
3. Calculates moisture percentage.
4. Displays information on OLED.
5. Controls relay operation.
6. Executes irrigation logic.

Why ESP32?

ESP32 was selected because of:

- Low cost
- High performance
- IoT compatibility
- Large community support

3.6 Output Layer

3.6.1 OLED Display

The OLED display provides real-time visual feedback to users.

Displayed Parameters

- Temperature
- Humidity
- Soil Moisture Percentage
- Pump Status

Communication Protocol

The OLED communicates with ESP32 using the I2C protocol.

Pin Connections

SDA → GPIO21

SCL → GPIO22

Advantages

- Low power consumption
- Compact size
- High contrast display
- Easy integration

3.7 Irrigation Control Layer

3.7.1 Relay Module

The relay module acts as an electrical switch controlled by ESP32.

Functions

- Isolates ESP32 from pump circuit
- Switches water pump ON/OFF
- Enables automatic irrigation

Relay Configuration

Relay Type: Active LOW

Control Pin: GPIO26

Working

When ESP32 outputs LOW signal:

- Relay activates
- Pump turns ON

When ESP32 outputs HIGH signal:

- Relay deactivates
- Pump turns OFF

3.7.2 Water Pump

The water pump provides irrigation water to plants.

Specifications

- Operating Voltage: 3V–9V
- Low Power Consumption
- Suitable for mini irrigation systems

Purpose

The pump supplies water whenever soil moisture falls below the predefined threshold.

3.8 Data Flow in the System

The flow of information follows the sequence below:

Step 1: DHT11 measures temperature and humidity.

Step 2: Soil moisture sensor measures soil moisture.

Step 3: Sensor readings are transmitted to ESP32.

Step 4: ESP32 processes sensor data.

Step 5: Moisture percentage is calculated.

Step 6: OLED display is updated.

Step 7: Moisture value is compared with threshold.

Step 8: Relay status is determined.

Step 9: Pump is switched ON or OFF.

Step 10: Cycle repeats continuously.

3.9 System Working Flow

The operational sequence of the system is:

START

↓

Initialize Sensors

↓

Read Temperature

↓

Read Humidity

↓

Read Soil Moisture

↓

Calculate Moisture Percentage

↓

Display Sensor Data

↓

Check Moisture Threshold

↓

Is Moisture Below Threshold?

↓

YES → Turn Pump ON

NO → Keep Pump OFF

↓

Repeat Process

3.10 Advantages of Proposed Architecture

The architecture provides several benefits:

- Modular Design
- Easy Maintenance
- Low Cost
- Real-Time Monitoring
- Expandable System
- Efficient Water Usage
- Suitable for IoT Applications

The modular structure allows additional sensors and cloud services to be integrated in future versions.

3.11 Summary

This chapter presented the architecture of the IoT Smart Irrigation and Plant Monitoring System. The system consists of sensors for data acquisition, ESP32 for processing, OLED for visualization, and relay-controlled pump for irrigation. The architecture enables real-time monitoring and automated irrigation, providing an efficient solution for smart agriculture applications.